

University College of Science Tumkur



SEMINAR REPORT

TOPIC:

VITAMINS

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An Overview of Vitamins:

Vitamins are organic compounds that are essential in small amounts for supporting normal physiological function. Since the human body either does not produce enough of them or does not produce them at all, vitamins must be obtained primarily through diet. They play a critical role in a wide range of bodily functions, including immune response, energy production, and bone health.



1. Fat-Soluble Vitamins

Fat-soluble vitamins are absorbed with dietary fats and are stored in the liver and fatty tissues for later use. Because they can accumulate in the body, consuming them in excessive amounts can lead to toxicity.

Vitamin A (Retinol):

Function: Supports vision, immune function, reproduction, and cellular communication.

Sources: Liver, dairy products, and fish; plant-based sources include beta-carotene found in carrots, sweet potatoes, and leafy greens.

Deficiency: Can lead to night blindness and a weakened immune system.

Toxicity: Excess intake, particularly from supplements, may cause liver damage and birth defects.



Vitamin D (Calciferol):

Function: Facilitates calcium absorption, promotes bone growth and remodelling.

Sources: Sunlight exposure, fortified dairy products, egg yolks, and fatty fish.

Deficiency: Leads to rickets in children and osteocalcin or osteoporosis in adults.

Toxicity: Can cause hypercalcemia, leading to nausea, weakness, and kidney issues.



Vitamin E (Tocopherol):

Function: Acts as an antioxidant, protecting cells from damage caused by free radicals.

Sources: Nuts, seeds, vegetable oils, and green leafy vegetables.

Deficiency: Rare, but can cause nerve and muscle damage, vision problems.

Toxicity: High doses may interfere with blood clotting and increase the risk of bleeding.



Vitamin K:

Function: Essential for blood clotting and bone metabolism.

Sources: Leafy greens (like spinach and kale), broccoli, and Brussels sprouts.

Deficiency: Can result in excessive bleeding and easy bruising.

Toxicity: Rare, but synthetic forms can lead to liver damage when taken in large amounts.



2. Water-Soluble Vitamins

Unlike fat-soluble vitamins, water-soluble vitamins are not stored in large amounts in the body and are usually excreted in urine. They need to be consumed more regularly through food.

Vitamin C (Ascorbic Acid):

Function: Important for the growth and repair of tissues, antioxidant, enhances iron absorption.

Sources: Citrus fruits, strawberries, bell peppers, broccoli, and tomatoes.

Deficiency: Causes scurvy, which leads to bleeding gums, bruising, and poor wound healing.

Toxicity: Large doses may cause stomach upset and kidney stones.



B Vitamins:

These vitamins are critical for energy metabolism and red blood cell formation. Each plays a unique role:

B1 (Thiamine):

Helps convert nutrients into energy.

Sources: Whole grains, pork, legumes.

Deficiency: Beriberi, characterized by weakness and nerve damage.

B2 (Riboflavin):

Supports energy production and skin health.

Sources: Milk, eggs, green vegetables.

Deficiency: Causes mouth sores and skin disorders.

B3 (Niacin):

Aids in digestion, skin health, and nervous system function.

Sources: Meat, fish, whole grains.

Deficiency: Pellagra (symptoms include diarrhoea, dermatitis, and dementia).

B5 (Pantothenic Acid):

Necessary for synthesizing coenzyme-A.

Sources: Almost all foods, including meat, eggs, and vegetables.

Deficiency: Rare; may cause fatigue and irritability.

B6 (Pyridoxine):

Important for protein metabolism and cognitive development.

Sources: Fish, poultry, bananas, potatoes.

Deficiency: Can cause anaemia and neurological symptoms.

B7 (Biotin):

Crucial for fat and carbohydrate metabolism.

Sources: Eggs, nuts, and whole grains.

Deficiency: Rare; symptoms include hair thinning and skin rashes.

B9 (Folate or Folic Acid):

Important for DNA synthesis and during pregnancy.

Sources: Leafy greens, legumes, and fortified cereals.

Deficiency: Leads to anaemia and birth defects.

B12 (Cobalamin):

Vital for nerve function and red blood cell formation.

Sources: Animal products like meat, dairy, and eggs.

Deficiency: Can cause pernicious anaemia and neurological issues.



Importance of a Balanced Vitamin Intake:

A well-balanced diet typically provides all the essential vitamins. However, certain populations—such as pregnant women, elderly individuals, vegetarians, and people with medical conditions—may require supplements. It's crucial to consult healthcare providers before starting vitamin supplements to avoid adverse effects.

Vitamin Supplementation: Pros and Cons

Pros: Can correct deficiencies

Beneficial during pregnancy (e.g., folic acid)

Helpful for individuals with restricted diets

Cons:

Risk of overdose, particularly with fat-soluble vitamins

Supplements are not a substitute for a healthy diet

Some may interact with medications

Conclusion:

Vitamins are indispensable to human health, affecting everything from growth and immune function to mental clarity and cellular repair. While modern diets often provide adequate levels, awareness of specific needs based on lifestyle, age, and health conditions is essential. The key is balance—too little can lead to deficiencies, and too much can result in toxicity. A diet rich in fruits, vegetables, whole grains, and lean proteins is the most effective way to maintain optimal vitamin intake and overall health.